

1.) See MATLAB code below for calculations.

Answers:

a) $V_{PFR} = 71.93$; $V_{CSTR} = 24.00$

b) Volumes are equal when X is between 0.40 and 0.60

c) $X = 0.752$

d) $PFR\ X = 0.401$; $CSTR\ X = 0.681$

e) $CSTR\ X = 0.40$; $PFR\ X = 0.90$

2.) a) Fireflies: $E_a = 48.2\text{ kJ/mol}$; $R^2 = 0.9952$

Crickets: $E_a = 51.3\text{ kJ/mol}$; $R^2 = 0.9999$

Ants: $E_a = 64.6\text{ kJ/mol}$; $R^2 = 0.9673$

Honeybees: $E_a = 111.3\text{ kJ/mol}$; $R^2 = 0.9744$

b) $v(40C) = 6.436\text{ cm/s}$

$v(-5C) = 4.9201e-03\text{ cm/s}$

c) All four follow the Arrhenius equation, when $\ln k$ is plotted against $1/T_m$ the point falls close to the straight lines with R^2 above 0.99. This indicates that their flight speed is temperature-dependent and exponential. Fireflies and crickets have similar E_a values, suggesting similar temperature sensitivity. Ants have a slightly higher E_a and honeybees have the highest E_a , meaning honeybees have the most temperature-dependent flight speeds.

d) More data would strengthen the relationships. Per the data provided, each conclusion is based on only 3 or 4 temperature points, so measurement error would produce a noticeable change. I would propose collecting data across a wider temperature range with smaller increments to further solidify the Arrhenius model.

e1) $v(41.5C) = 19.161\text{ cm/s}$

e2) We could apply linearization to make the problem more simple. If we take the natural log of the speed and plot it against $1/T$, the Arrhenius equation becomes a straight line for simple linear regression.

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clear; clc; close all;

% Parameters
FA0 = 300;
Xtab = [0 0.20 0.40 0.45 0.50 0.60 0.80 0.90];
rneg = [1.00 1.67 5.00 5.00 5.00 5.00 1.25 0.91];
invR = 1./rneg;

% Cumulative PFR volume vs X
Vcum = zeros(size(Xtab));
for i = 2:numel(Xtab)
    dX = Xtab(i) - Xtab(i-1);
    avg = 0.5*(invR(i) + invR(i-1));
    Vcum(i) = Vcum(i-1) + FA0*avg*dX;
end

% linear interpolation for Y(X) when X is between tab points
interpY = @(x, Xv, Yv) ...
    ( ...
        (x <= Xv(1)) .* Yv(1) + ...
        (x >= Xv(end)) .* Yv(end) + ...
        (x > Xv(1) & x < Xv(end)) .* ( ...
            Yv(arrayfun(@(xx)find(Xv<=xx,1,'last'), x)) + ...
            ( (x - Xv(arrayfun(@(xx)find(Xv<=xx,1,'last'), x))) ./ ...
                (Xv(arrayfun(@(xx)find(Xv>xx,1,'first'), x)) -
                Xv(arrayfun(@(xx)find(Xv<=xx,1,'last'), x))) ) .* ...
            (Yv(arrayfun(@(xx)find(Xv>xx,1,'first'), x)) -
            Yv(arrayfun(@(xx)find(Xv<=xx,1,'last'), x))) ) ...
        ) ...
    );

% Part a
Xtarget = 0.40;
invR_at_04 = interp1(Xtab, invR, Xtarget, 'linear');
r_at_04 = 1/invR_at_04;

% PFR volume
V_PFR_04 = interp1(Xtab, Vcum, Xtarget, 'linear');

% CSTR volume
V_CSTR_04 = FA0 * Xtarget / r_at_04;

% Part b
% From the table, -rA = 5 between X=0.40 and 0.60

% Part c
V_CSTR_goal = 105;

Xgrid = linspace(0,0.90,5001);
rgrid = interp1(Xtab, rneg, Xgrid, 'linear');
Vcstr = FA0 .* (Xgrid - 0)./ rgrid;

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[~,iBest] = min(abs(Vcstr - V_CSTR_goal));
X_CSTR_105 = Xgrid(iBest);

% Part d:
V1 = 72; V2 = 24;

Xfine = linspace(0,0.90,5001);
Vfine = interp1(Xtab, Vcum, Xfine, 'linear');

[~,i1] = min(abs(Vfine - V1));
X1 = Xfine(i1);

X2grid = linspace(X1,0.90,5001);
r2grid = interp1(Xtab, rneg, X2grid, 'linear');
Vcstr_2 = FA0.*(X2grid - X1)./r2grid;

[~,jBest] = min(abs(Vcstr_2 - V2));
X2_pfr_then_cstr = X2grid(jBest);

% Part e
V1 = 24; V2 = 72;

X1grid = linspace(0,0.90,5001);
r1grid = interp1(Xtab, rneg, X1grid, 'linear');
Vcstr_1 = FA0.*(X1grid - 0)./r1grid;
[~,kBest] = min(abs(Vcstr_1 - V1));
X1_cstr = X1grid(kBest);

% PFR from X1 to X2
Vcum_fine = FA0 * cumtrapz(Xfine, 1./interp1(Xtab, rneg, Xfine, 'linear'));
V_X1 = interp1(Xfine, Vcum_fine, X1_cstr, 'linear');
X2grid = linspace(X1_cstr, 0.90, 5001);
V_X2 = interp1(Xfine, Vcum_fine, X2grid, 'linear');
Vpfr_between_vals = V_X2 - V_X1;

[~,mBest] = min(abs(Vpfr_between_vals - V2));
X2_cstr_then_pfr = X2grid(mBest);

% Part f
Vgrid = linspace(0,100,400);
% Invert Vcum_fine(X) -> X(V) by interp1 swapping axes:
X_of_V = interp1(Vcum_fine, Xfine, Vgrid, 'linear', 'extrap');
r_of_V = interp1(Xtab, rneg, X_of_V, 'linear', 'extrap');

figure; plot(Vgrid, X_of_V, 'LineWidth', 1.5);
xlabel('PFR volume V (dm^3)'); ylabel('Conversion X'); grid on;
title('Q1(f) - X vs V');

figure; plot(Vgrid, r_of_V, 'LineWidth', 1.5);
xlabel('PFR volume V (dm^3)'); ylabel('-r_A (mol/dm^3/min)'); grid on;
title('Q1(f) - -r_A vs V');

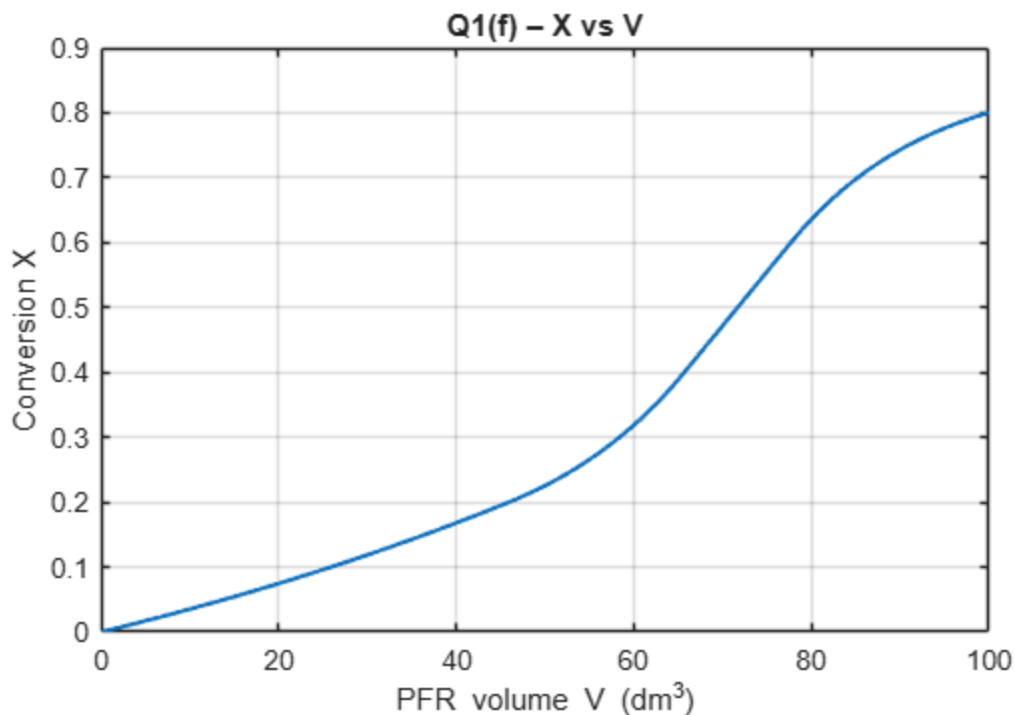
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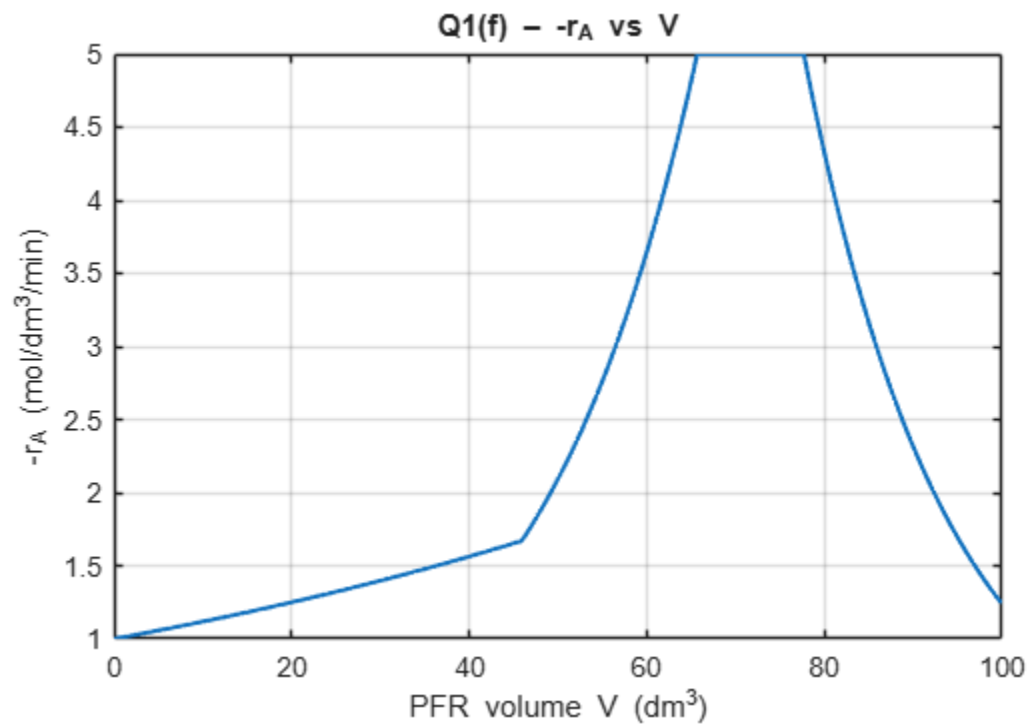
% Answers

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fprintf('\nAnswers:\n');  
fprintf(' (a) V_PFR(40%)= %.2f dm^3, V_CSTR(40%)= %.2f dm^3\n', V_PFR_04,  
V_CSTR_04);  
fprintf(' (b) Equal volumes when 0.40<=X<=0.60\n');  
fprintf(' (c) X in 105-dm^3 CSTR: %.3f\n', X_CSTR_105);  
fprintf(' (d) 72 PFR -> X= %.3f; then +24 CSTR -> X= %.3f\n', X1,  
X2_pfr_then_cstr);  
fprintf(' (e) 24 CSTR -> X= %.3f; then +72 PFR -> X= %.3f\n\n', X1_cstr,  
X2_cstr_then_pfr);
```

Answers:

- (a) $V_{PFR}(40\%) = 71.93 \text{ dm}^3$, $V_{CSTR}(40\%) = 24.00 \text{ dm}^3$
- (b) Equal volumes when $0.40 \leq X \leq 0.60$
- (c) X in 105-dm^3 CSTR: 0.752
- (d) $72 \text{ PFR} \rightarrow X = 0.401$; then $+24 \text{ CSTR} \rightarrow X = 0.681$
- (e) $24 \text{ CSTR} \rightarrow X = 0.400$; then $+72 \text{ PFR} \rightarrow X = 0.900$





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clear; clc; close all;

R = 8.314;    % J/mol-K

% Data
Tf = [21 25 30];      kf = [9.0 12.16 16.2];      % Fireflies
Tc = [14.2 20.3 27.0]; kc = [80 126 200];        % Crickets
Ta = [10 20 30 38];   ka = [0.5 2.0 3.4 6.5];     % Ants
Th = [25 30 35];      kh = [0.7 1.8 3.0];        % Honeybees

% Fireflies
x_f = 1./(Tf + 273.15); y_f = log(kf);
p_f = polyfit(x_f, y_f, 1);          % y = m x + b
m_f = p_f(1); b_f = p_f(2);
Ea_f = -m_f*R; A_f = exp(b_f);
yhat_f = polyval(p_f, x_f);
R2_f = 1 - sum((y_f - yhat_f).^2)/sum((y_f - mean(y_f)).^2);

% Crickets
x_c = 1./(Tc + 273.15); y_c = log(kc);
p_c = polyfit(x_c, y_c, 1);
m_c = p_c(1); b_c = p_c(2);
Ea_c = -m_c*R; A_c = exp(b_c);
yhat_c = polyval(p_c, x_c);
R2_c = 1 - sum((y_c - yhat_c).^2)/sum((y_c - mean(y_c)).^2);

% Ants
x_a = 1./(Ta + 273.15); y_a = log(ka);
p_a = polyfit(x_a, y_a, 1);
m_a = p_a(1); b_a = p_a(2);
Ea_a = -m_a*R; A_a = exp(b_a);
yhat_a = polyval(p_a, x_a);
R2_a = 1 - sum((y_a - yhat_a).^2)/sum((y_a - mean(y_a)).^2);

% Honeybees
x_h = 1./(Th + 273.15); y_h = log(kh);
p_h = polyfit(x_h, y_h, 1);
m_h = p_h(1); b_h = p_h(2);
Ea_h = -m_h*R; A_h = exp(b_h);
yhat_h = polyval(p_h, x_h);
R2_h = 1 - sum((y_h - yhat_h).^2)/sum((y_h - mean(y_h)).^2);

% Part a
fprintf('\na: Activation energies and R^2\n');
fprintf(' Fireflies : Ea = %.1f kJ/mol, R^2 = %.4f\n', Ea_f/1000, R2_f);
fprintf(' Crickets  : Ea = %.1f kJ/mol, R^2 = %.4f\n', Ea_c/1000, R2_c);
fprintf(' Ants       : Ea = %.1f kJ/mol, R^2 = %.4f\n', Ea_a/1000, R2_a);
fprintf(' Honeybees : Ea = %.1f kJ/mol, R^2 = %.4f\n', Ea_h/1000, R2_h);

% Part b
T40K = 40 + 273.15; Tm5K = -5 + 273.15;
v40 = A_h * exp(-Ea_h/(R*T40K));

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v_m5 = A_h * exp(-Ea_h/(R*Tm5K));
fprintf('\nb: Honeybee\n');
fprintf(' v(40 C) = %.3f cm/s\n', v40);
fprintf(' v(-5 C) = %.4e cm/s\n', v_m5);

% Part c: see write up
% Part d: see write up

% Part e1
Tb = [27 37 40]; kb = [6.5 13 18];
x_b = 1./(Tb + 273.15); y_b = log(kb);
p_b = polyfit(x_b, y_b, 1);
m_b = p_b(1); b_b = p_b(2);
Ea_b = -m_b*R; A_b = exp(b_b);
T415K = 41.5 + 273.15;
v415 = A_b * exp(-Ea_b/(R*T415K));
fprintf('\ne1: Beetle\n');
fprintf(' v(41.5 C) = %.3f cm/s\n\n', v415);

a: Activation energies and R^2
  Fireflies : Ea = 48.2 kJ/mol, R^2 = 0.9952
  Crickets  : Ea = 51.3 kJ/mol, R^2 = 0.9999
  Ants      : Ea = 64.6 kJ/mol, R^2 = 0.9673
  Honeybees : Ea = 111.3 kJ/mol, R^2 = 0.9744

b: Honeybee
  v(40 C) = 6.436 cm/s
  v(-5 C) = 4.9201e-03 cm/s

e1: Beetle
  v(41.5 C) = 19.161 cm/s

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